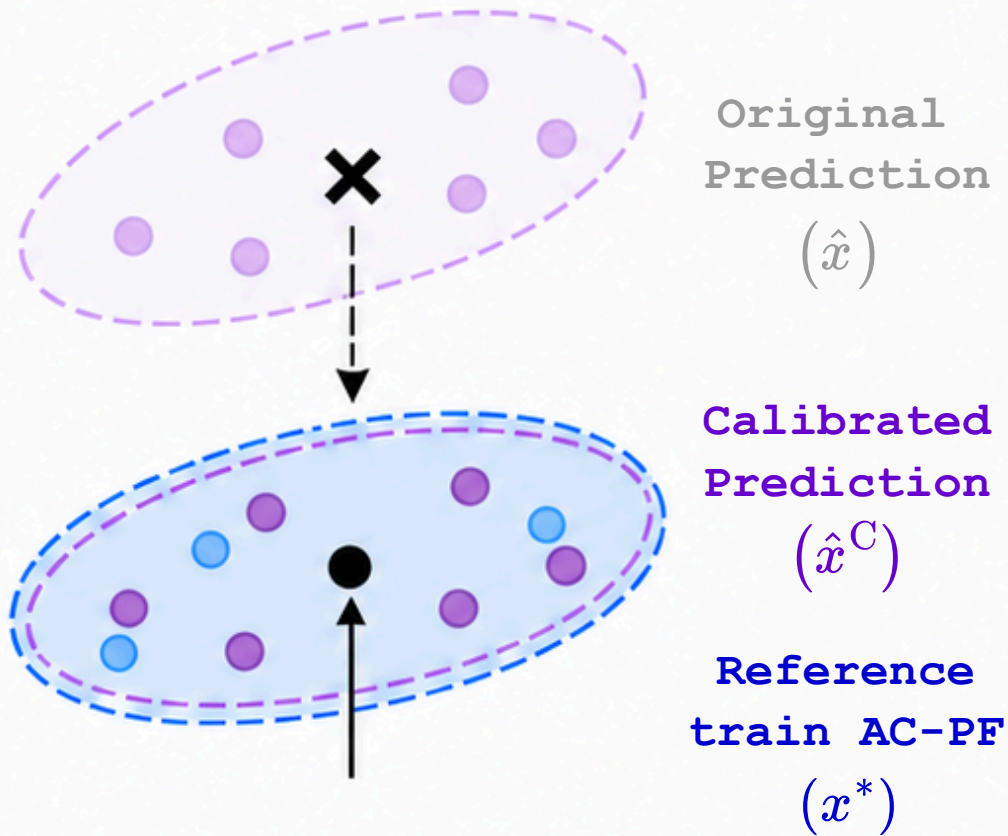


Reduced Small voltage
Error and Residual

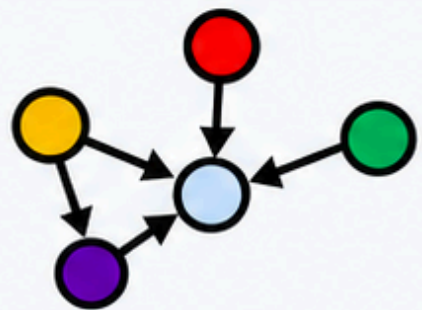
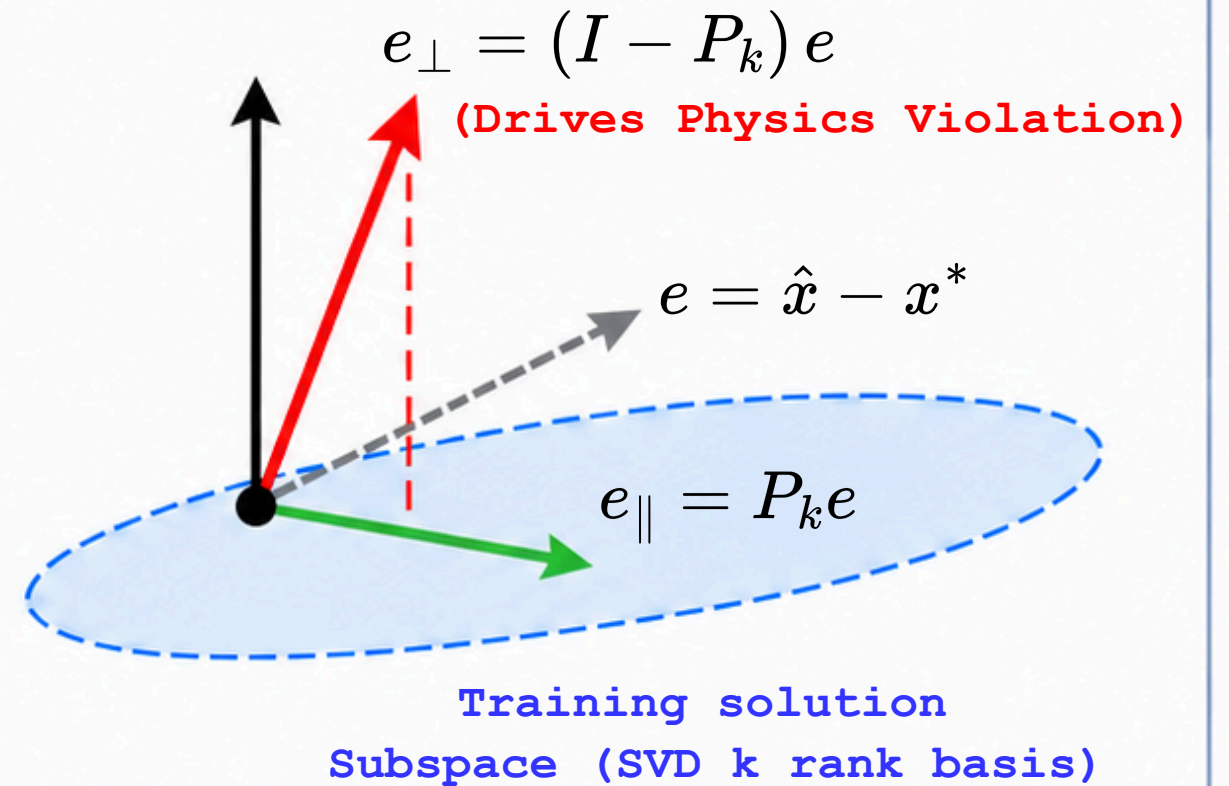
$$\|\hat{x}^{\text{CSP}} - x^*\| \quad \text{small}$$

$$\|r(\hat{x}^{\text{CSP}}; u)\| \quad \text{small}$$

1. Bias Calibration ($\hat{b}$)

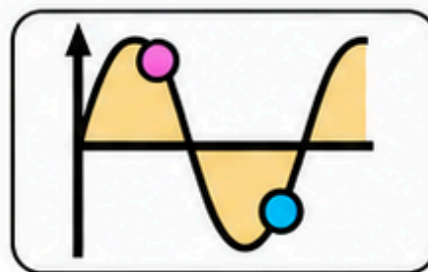


2. Subspace Projection (P_k)



$$r(\hat{x}; u) = S_{\text{set}}(u) - \hat{V} \odot (Y\hat{V})^*$$

Physics Informed



PIGNN
GridSFM
gridfm
LUMINA

$$\mathcal{L} = \lambda_{\text{sup}} \|\hat{x} - x^*\|^2 + \lambda_{\text{phys}} (\|\Delta P\|^2 + \|\Delta Q\|^2)$$

Small Voltage Error But Large Residual

$$\|\hat{x} - x^*\| \quad \text{small}$$

$$\|r(\hat{x}; u)\| \quad \text{large}$$